

Ruihao Xi

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Shanghai, China

EDUCATION

• Tongji University

Sep. 2022 – Jul. 2026

B.S. in Computer Science and Technology

Shanghai, China

◦ **GPA:** 4.84/5.00 | **Grade:** 93.37/100 | **Rank:** 2/52

◦ Relevant Coursework:

- * Research Focus: Principles and Techniques of Artificial Intelligence; Pattern Recognition; Neural Network and Deep Learning;
- * Advanced Computer Science Foundations: Advanced Programming Languages; Principles of Compiler Design; Computer System Architecture
- * CS 188 Introduction to Artificial Intelligence (UC Berkeley); CS 15-445: Database Systems (CMU)

◦ Major Accomplishments:

- * **National Scholarship for Undergraduate Students (2022-2023):** Top 0.2 % nationwide; China's highest undergraduate government scholarship.
- * **Shanghai Municipal Government Scholarship for Undergraduate Students (2023-2024):** Awarded to 1-2 students per college annually; provincial-level recognition.
- * **Joyce M. Kuok Foundation Excellent Scholarship (2024-2025):** Awarded to 1-2 students per college annually;
- * **Outstanding Student | Tongji University (2022-2023; 2023-2024) :** Top 5% of students annually; awarded based on academic and extracurricular excellence.

RESEARCH INTERESTS

Multimodal learning with a focus on vision as a foundation for reasoning and knowledge acquisition; Developing **unified models** for integrating vision, language, and other modalities; Exploring **visual understanding** for complex reasoning, causality, and **cross-modal learning**.

PROJECTS EXPERIENCE

• Real-Time Vision System for Robotics

October 2024 – Current

C++, OpenVino, YOLO, MPC

 (190+ Stars)]

- Designed and implemented automated targeting and shooting mechanisms for robotic vehicles, focusing on real-time object detection and tracking to enhance operational performance.
- Developed and deployed a YOLO-based image recognition model on a custom dataset of armored plates, utilizing C++ and OpenVino for deployment.
- Addressed environmental challenges (lighting variations, observer movement, motion control errors) through ROI optimization, yaw adjustment, and bitmap-based firing logic.
- Led cross-functional collaboration: reduced rotational inertia with mechanical team, implemented latency command compensation with electrical team.
- Enhanced real-time decision-making capabilities by integrating feedback loops and adaptive algorithms, ensuring robust performance in dynamic environments.
- Achieved multiple top-tier awards, including:
 - * RoboMaster 2025 Super Confrontation:
 - Champion (Eastern Region); National Top 8 (National Final)
 - Ranked #1 in overall auto-aim accuracy (Eastern Region).
 - Team robots (Sentry & Standard) earned 10+ MVP awards for performance.
 - * RoboMaster 2025 University League:
 - 2nd Runner-up (3rd Place) (Shanghai Regional)

• Text-Driven 2D Motion Generation

Jul. 2024 – Jul. 2025

Python

- Developed a model to generate 2D skeletal animations from textual descriptions, focusing on enhancing motion realism and adaptability across diverse input scenarios.

- Proposed a novel dataset for two-person interaction scenarios, addressing the gap in existing text-driven motion generation research that primarily focuses on single-person 3D motion.
- Curated and preprocessed the dataset by combining publicly available datasets with web-scraped data, followed by annotation using RTMPose and Owl3. Conducted rigorous data cleaning to remove inaccuracies, misidentified frames, and incomplete skeletal data.
- Designed and implemented a Transformer-based model with a custom attention mechanism to improve motion generation quality. Integrated deep reinforcement learning strategies to further optimize the model's performance.
- Contributed to advancing research in text-driven motion generation by addressing limitations in existing datasets and methodologies, paving the way for more realistic and interactive motion synthesis.
- Authored a co-first author manuscript, "Toward Rich Video Human-Motion2D Generation" detailing this work, which is currently under review at the CVPR 2026 Conference.

• Mandelbrot Fractal Antenna Design

Feb. 2023 – Apr. 2024

Python, MatLab

- Led automation modeling and simulation programming for antenna design, leveraging a combination of MATLAB, Visual Basic, and Python to enhance traditional algorithms and optimize design parameters.
- Developed a tile-based strategy in MATLAB to improve the escape-time algorithm, enabling high-precision generation of fractal structures and multi-connected fractal geometries.
- Automated simulation workflows by integrating Python with HFSS for parametric optimization and design validation, streamlining the antenna design process.
- Explored advanced design methodologies for high-profile Mandelbrot fractal antennas, contributing to innovative research in the field.
- Published two first-authored research papers:
 - * A Multiband Microstrip Patch Antenna with Mandelbrot Fractal Structure at <conf>
 - * An Automatized Simulation Method for Design of Mandelbrot Fractal Antenna at <conf>
- Secured a patent for the developed antenna design (Patent Application No.: 202311796896.4).

PROFESSIONAL EXPERIENCE

• Zhijing Technology [🌐]

September 2025 - November 2025

Generative AI Intern

Shanghai, China

- Developed a semantic-aware ranking model to address the "copy-paste" (data memorization) issue in the generative model, achieving 96% accuracy on the validation set.
- Integrated this model as a core reward function within an RL framework to optimize the generative model's output diversity and quality.
- Optimized the FlowGRPO algorithm by innovating its RL reward logic, inspired by Direct Preference Optimization (DPO).
- Redefined the 'lose-win' case classification, enabling the model to reach target performance 3x faster.

• Oxford Suzhou Centre for Advanced Research [🌐]

August 2025 - August 2025

Summer Research Intern

Suzhou, China

- Developed an AI-based image recognition solution to classify cells from microscope images, outputting statistical results while rejecting debris, overlapping, and partial cells.
- Authored high-reliability, well-commented Python code and a step-by-step user guide; the solution is still actively used by team members.
- Conceptualized a distributed hardware and software architecture for simultaneous control and monitoring of multiple automated cell bioreactors, focusing on reliability, stability, and scalability.

PATENTS AND PUBLICATIONS

UR=UNDER REVIEW, C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

[UR] Ruihao Xi, et al. (2025). **Toward Rich Video Human-Motion2D Generation**. In *IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR 2026)*.

[C.2] Ruihao Xi, et al. (2024). **An Automatized Simulation Method for Design of Mandelbrot Fractal Antenna**. In *International Conference on Microwaves, Communications, Antennas and Electronic Systems (IEEE COMCAS 2024)*. DOI: 10.1109/COMCAS58210.2024.10666225

[C.1] Ruihao Xi, et al. (2023).**A Multiband Microstrip Patch Antenna with Mandelbrot Fractal Structure**. In *Photonics and Electromagnetics Research Symposium (PIERS 2023)*. DOI: 10.1109/PIERS59004.2023.10221535

[P.1] Ruihao Xi, et al. (2023). **Mandelbrot Fractal Antenna Design**. Patent Office. Patent Application No. 202311796896.4. Status: Patent Pending.

AWARDS AND COMPETITIONS

• Gold Award (0.02% - state level) ,China International College Students’ Innovation and Entrepreneurship Competition	2024
• Second Prize (approx. 10% - state level) , The Contemporary Undergraduate Mathematical Contest in Modeling	2024
• Honorable Mention (approx. 21% - international) , 2024 Mathematical Contest in Modeling (MCM/ICM, USA)	2024
• “Science & Technology Rising Star” Award , 22nd Shanghai International Youth Science & Technology Competition Innovation Contest	2024
• "My Favorite Project" Award , The 2024 National Undergraduate Innovation Conference	2023
• First Prize ,The 18th Challenge Cup National College Students’ Extracurricular Academic Science and Technology Contest (High-Tech Track)	2023
• First Prize (< 8%) , 15th Shanghai Municipal College Student Mathematics Contest (Higher Education Press Cup)	2023
• First Prize (< 8%) , 15th National College Student Mathematics Contest	2023

SKILLS

- **Programming Languages:** Python(Expert), C++(Expert), Java(Intermediate), MATLAB(Intermediate)
- **AI/ML Frameworks:** PyTorch(Expert), related deep learning libraries(Advanced)
- **Tools & Environments:** Git/GitHub(Advanced), VSCode(Advanced), Linux(Advanced), Anaconda(Advanced)
- **Other:** MySQL(Intermediate), FPGA development(Intermediate), web deployment(Intermediate), network configuration(Intermediate)